

Weld Schedule Manager (Root Pass)



WPS No	V6	Position / Direction	Fixed 5G - Down	Diameter (in)	48	Wall (mm)	15.88
Process	LSC (Modified Short Arc)	Specification	API1104	Filler Wire	Lincoln S6	Shield Gas	Ar/CO2 85/15
Bevel Design	30 degree	Material Grade	X70	Trade Name	Sure-arc	Completed by	Ben Hatch

0°	Travel Speed	Torch Angle	Dynamic Correction
0mm	14	-20	1
1mm	12	5	0
2mm	7	5	0
3mm	4.5	5	0
4mm	3	5	0
5mm	2	5	0

30°	Travel Speed	Torch Angle	Dynamic Correction
0mm	15	-15	0
1mm	13	0	0
2mm	9	0	0
3mm	6.5	0	0
4mm	4.5	-5	0
5mm	2.75	-5	0

60°	Travel Speed	Torch Angle	Dynamic Correction
0mm	15.5	-10	-1
1mm	13.5	-5	0
2mm	10	-10	0
3mm	7.5	-15	0
4mm	5	-15	0
5mm	2.75	-15	0

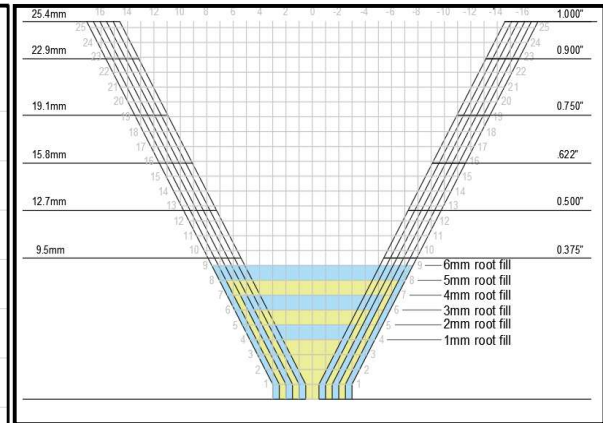
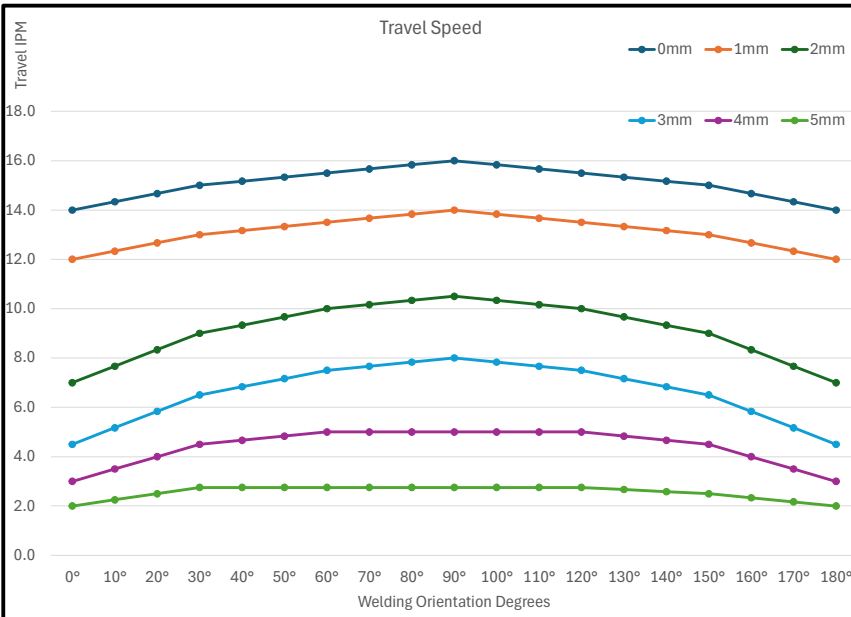
90°	Travel Speed	Torch Angle	Dynamic Correction
0mm	16	-5	-2
1mm	14	-5	0
2mm	10.5	-10	0
3mm	8	-15	0
4mm	5	-15	0
5mm	2.75	-15	0

120°	Travel Speed	Torch Angle	Dynamic Correction
0mm	15.5	0	-3
1mm	13.5	-5	0
2mm	10	-10	0
3mm	7.5	-15	0
4mm	5	-15	0
5mm	2.75	-15	0

180°	Travel Speed	Torch Angle	Dynamic Correction
0mm	14	0	-5
1mm	12	0	0
2mm	7	0	0
3mm	4.5	0	0
4mm	3	0	0
5mm	2	0	0

150°	Travel Speed	Torch Angle	Dynamic Correction
0mm	15	0	-4
1mm	13	0	0
2mm	9	-5	0
3mm	6.5	-15	0
4mm	4.5	-15	0
5mm	2.5	-15	0

Deposition Adjustment			CTWD	Arc Length	Weave Type	Weave Frequency	Weave Width	Weave Length	Weave Dwell
Gap	Whole Pipe	Lower Half							
0mm	-10%	0%	0	-4	None	0	1.2	1.2	0
1mm	0%	0%	0	-3	None	0	1.2	1.2	0
2mm	0%	0%	1	-3	Circle	12	2.2	2.2	0
3mm	0%	0%	2	0	Circle	12	3.6	3.6	0
4mm	0%	0%	2	0	Circle	12	5.2	3.8	0
5mm	10%	10%	3	0	Circle	12	6.4	4.8	0



Cleaning / Grinding				
Root Gap	Grind Weld Pass	Travel Speed	Weave Amplitude	Frequency Hz
0mm	ON	75	0.5	2
1mm	ON	70	1	2
2mm	ON	60	2	2
3mm	ON	50	3	2
4mm	ON	50	4	2
5mm	ON	30	5	2

Notes

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PIPE-CELL

Travel Speed IPM																			
Angle Degrees >	0°	10°	20°	30°	40°	50°	60°	70°	80°	90°	100°	110°	120°	130°	140°	150°	160°	170°	180°
0mm	14.0	14.3	14.7	15.0	15.2	15.3	15.5	15.7	15.8	16.0	15.8	15.7	15.5	15.3	15.2	15.0	14.7	14.3	14.0
1mm	12.0	12.3	12.7	13.0	13.2	13.3	13.5	13.7	13.8	14.0	13.8	13.7	13.5	13.3	13.2	13.0	12.7	12.3	12.0
2mm	7.0	7.7	8.3	9.0	9.3	9.7	10.0	10.2	10.3	10.5	10.3	10.2	10.0	9.7	9.3	9.0	8.3	7.7	7.0
3mm	4.5	5.2	5.8	6.5	6.8	7.2	7.5	7.7	7.8	8.0	7.8	7.7	7.5	7.2	6.8	6.5	5.8	5.2	4.5
4mm	3.0	3.5	4.0	4.5	4.7	4.8	5.0	5.0	5.0	5.0	5.0	5.0	5.0	4.8	4.7	4.5	4.0	3.5	3.0
5mm	2.0	2.3	2.5	2.8	2.8	2.8	2.8	2.8	2.8	2.8	2.8	2.8	2.8	2.7	2.6	2.5	2.3	2.2	2.0

Wire Feed IPM																			
Angle Degrees >	0°	10°	20°	30°	40°	50°	60°	70°	80°	90°	100°	110°	120°	130°	140°	150°	160°	170°	180°
0mm	239	245	251	257	259	262	265	268	271	274	271	268	265	262	259	257	251	245	239
1mm	228	234	241	247	250	253	257	260	263	266	263	260	257	253	250	247	241	234	228
2mm	161	176	192	207	215	222	230	234	238	242	238	234	230	222	215	207	192	176	161
3mm	135	155	175	195	205	215	225	230	235	240	235	230	225	215	205	195	175	155	135
4mm	105	123	140	158	163	169	175	175	175	175	175	175	175	169	163	158	140	123	105
5mm	88	99	110	121	121	121	121	121	121	133	133	133	133	129	125	121	113	105	97

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Weave Frequency Hz																			
Angle Degrees >	0°	10°	20°	30°	40°	50°	60°	70°	80°	90°	100°	110°	120°	130°	140°	150°	160°	170°	180°
0mm	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1mm	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
2mm	1.40	1.53	1.67	1.80	1.87	1.93	2.00	2.03	2.07	2.10	2.07	2.03	2.00	1.93	1.87	1.80	1.67	1.53	1.40
3mm	0.90	1.03	1.17	1.30	1.37	1.43	1.50	1.53	1.57	1.60	1.57	1.53	1.50	1.43	1.37	1.30	1.17	1.03	0.90
4mm	0.60	0.70	0.80	0.90	0.93	0.97	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.97	0.93	0.90	0.80	0.70	0.60
5mm	0.40	0.45	0.50	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.53	0.52	0.50	0.47	0.43	0.40

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Process	LSC (Modified Short Arc)		Specification	API1104	Filler Wire	Lincoln S6	Shield Gas	Ar/C02 85/15
Bevel Design	30 degree		Material Grade	X70	Trade Name	Sure-arc	Completed by	Ben Hatch



4mm	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
5mm	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Contact to work distance (mm)																			
Angle Degrees >	0°	10°	20°	30°	40°	50°	60°	70°	80°	90°	100°	110°	120°	130°	140°	150°	160°	170°	180°
0mm	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1mm	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
2mm	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
3mm	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00
4mm	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00
5mm	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00

Arc Length																			
Angle Degrees >	0°	10°	20°	30°	40°	50°	60°	70°	80°	90°	100°	110°	120°	130°	140°	150°	160°	170°	180°
0mm	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4
1mm	-3	-3	-3	-3	-3	-3	-3	-3	-3	-3	-3	-3	-3	-3	-3	-3	-3	-3	-3
2mm	-3	-3	-3	-3	-3	-3	-3	-3	-3	-3	-3	-3	-3	-3	-3	-3	-3	-3	-3
3mm	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
4mm	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
5mm	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Voltage (WS * 0.07)																			
Angle Degrees >	0°	10°	20°	30°	40°	50°	60°	70°	80°	90°	100°	110°	120°	130°	140°	150°	160°	170°	180°
0mm	17	17	18	18	18	18	19	19	19	19	19	19	19	18	18	18	18	17	17
1mm	16	16	17	17	18	18	18	18	18	19	18	18	18	18	18	17	17	16	16
2mm	11	12	13	14	15	16	16	16	17	17	17	16	16	16	15	14	13	12	11
3mm	9	11	12	14	14	15	16	16	16	17	16	16	16	15	14	14	12	11	9
4mm	7	9	10	11	11	12	12	12	12	12	12	12	12	12	11	11	10	9	7
5mm	6	7	8	8	8	8	8	8	8	9	9	9	9	9	9	8	8	7	7

Amps (WS * 0.99)																			
Angle Degrees >	0°	10°	20°	30°	40°	50°	60°	70°	80°	90°	100°	110°	120°	130°	140°	150°	160°	170°	180°
0mm	237	243	248	254	257	260	262	265	268	271	268	265	262	260	257	254	248	243	237
1mm	226	232	238	245	248	251	254	257	260	263	260	257	254	251	248	245	238	232	226
2mm	159	175	190	205	213	220	228	231	235	239	235	231	228	220	213	205	190	175	159
3mm	134	153	173	193	203	213	223	228	233	238	233	228	223	213	203	193	173	153	134
4mm	104	121	139	156	162	167	173	173	173	173	173	173	173	167	162	156	139	121	104
5mm	87	98	109	120	120	120	120	120	120	132	132	132	132	128	124	120	112	104	96

Heat Input (kJ / in)																			
Angle Degrees >	0°	10°	20°	30°	40°	50°	60°	70°	80°	90°	100°	110°	120°	130°	140°	150°	160°	170°	180°
0mm	17	17	18	18	18	19	19	19	19	19	19	19	19	19	18	18	18	17	17
1mm	18	19	19	20	20	20	20	21	21	21	21	21	21	20	20	20	19	19	18
2mm	15	17	18	20	21	21	22	22	23	23	23	22	22	21	21	20	18	17	15
3mm	17	19	22	24	26	27	28	29	29	30	29	29	28	27	26	24	22	19	17
4mm	15	18	20	23	24	25	25	25	25	25	25	25	25	25	24	23	20	18	15
5mm	16	18	20	22	22	22	22	22	22	27	27	27	27	26	25	24	23	21	19

Heat Input (kJ / mm)																			
Angle Degrees >	0°	10°	20°	30°	40°	50°	60°	70°	80°	90°	100°	110°	120°	130°	140°	150°	160°	170°	180°
0mm	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.8	0.8	0.8	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7
1mm	0.7	0.7	0.7	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.7	0.7	0.7
2mm	0.6	0.7	0.7	0.8	0.8	0.8	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.8	0.8	0.8	0.7	0.7	0.6
3mm	0.7	0.8	0.9	1.0	1.0	1.1	1.1	1.1	1.2	1.2	1.2	1.1	1.1	1.1	1.0	1.0	0.9	0.8	0.7
4mm	0.6	0.7	0.8	0.9	0.9	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	0.9	0.9	0.8	0.7	0.6
5mm	0.6	0.7	0.8	0.9	0.9	0.9	0.9	0.9	0.9	1.1	1.1	1.1	1.1	1.0	1.0	1.0	0.9	0.8	0.8